

# Today's Agenda

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## 1 Tutorial Purpose

First 20 minutes: Building intuition and bridging the gap between statistical theory and practical applications

## 2 Attendance Policy

Canvas quiz with assigned student numbers released during break

**Important:** Technical issues must be reported to TA **immediately in person** — late email claims will not be accepted

## 3 Tutorial Structure

Remaining time dedicated to Stata coding practice to prepare for independent empirical projects

## 4 Session Policy

Please attend your assigned tutorial session

If you need to switch sessions or cannot attend, **email the TA in advance**

First 20 min: Theory & Intuition

Remaining time: Stata Practice

# ECO 375 | Statistical Review

## Appendices A–D — Top-Down Bridge

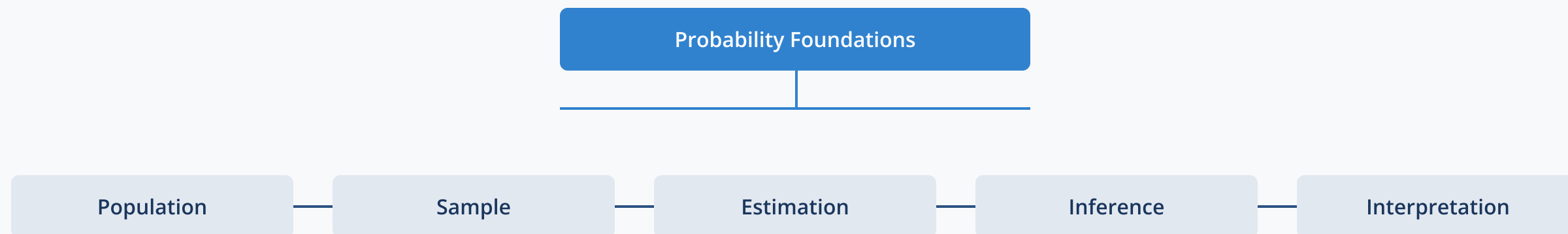
Why this review — and why now?

- Econometrics = Economics + Data + **Statistics**
- We use a **sample** to learn about a **population**
- Today: a big-picture **map** → then the essential tools



# The roadmap of econometrics

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- **Probability** provides essential infrastructure before we discuss population
- Econometric flow: Population → Sample → Estimation → Inference → Interpretation
- We need probability theory to define our target parameters in the population

# Where stats fit in the regression story

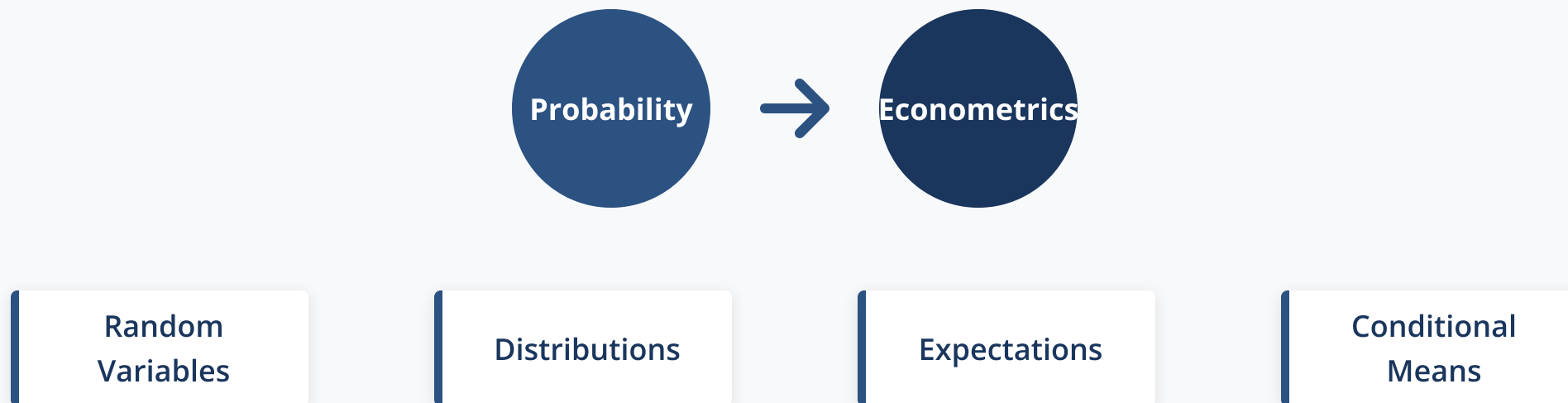
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- Coefficients are **statistics** from a sample
  - Standard errors quantify how much they would **wiggle** if we resampled
  - Tests & CIs are structured ways to handle **uncertainty**
  - Without this **statistical spine**, regression is just line-drawing

# Probability Foundations

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The essential infrastructure that powers all statistical inference — we must understand probability before we can build econometric models



# PDF vs CDF: The Building Blocks of Probability

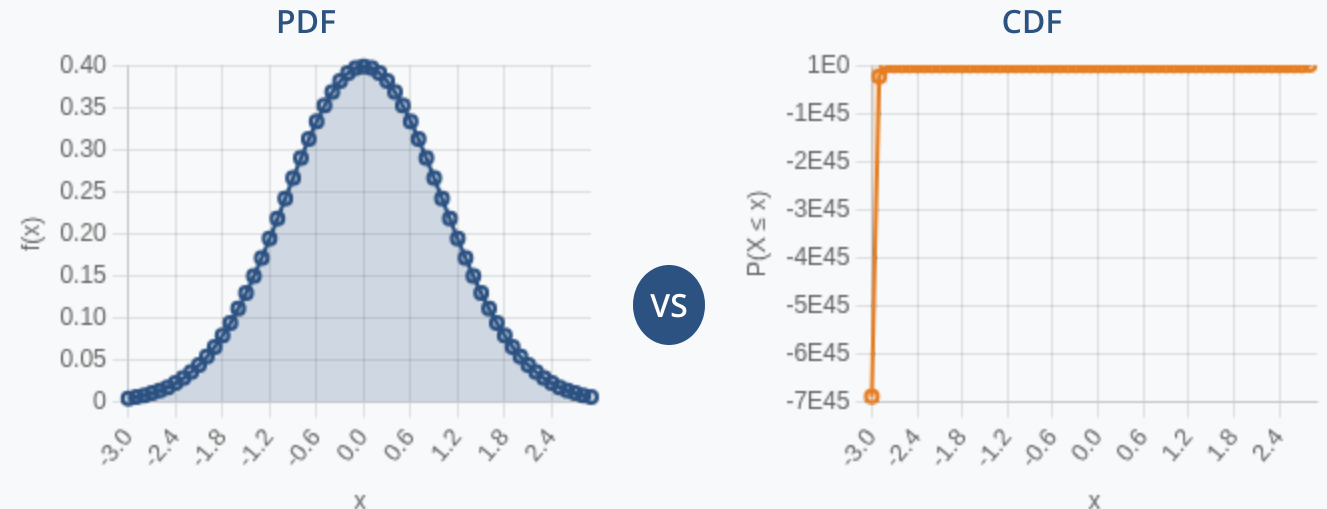
- **PDF (density function):** Height/likelihood at each value
- **CDF (cumulative):** Probability of  $X \leq x$
- **Z-standardization:** Transforms any normal to  $N(0,1)$
- Why standardize? **One table for all normal distributions**

$$Z = (X - \mu) / \sigma$$

If  $X \sim N(\mu, \sigma^2)$ , then  $Z \sim N(0,1)$

**Key insight:** Standard normal tables work for *any* normal distribution through standardization. This is how we'll compute p-values later!

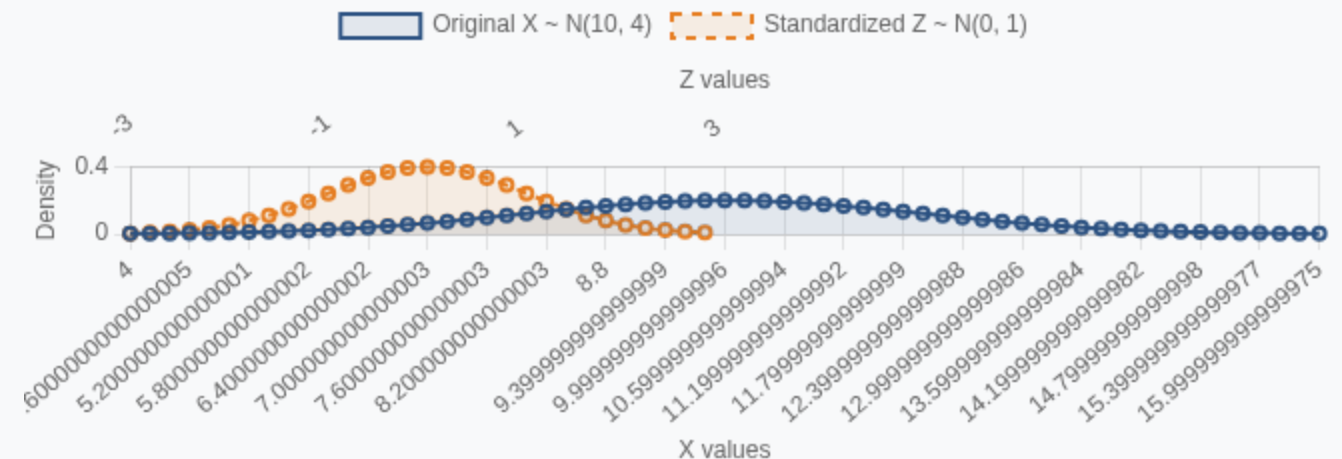
PDF vs CDF Comparison



Shows density at each point

Shows  $P(X \leq x)$

Z-Standardization: Any Normal  $\rightarrow$  Standard Normal



# Probability: describing the population

- Random variables, distributions, z-scores
- Tail areas **power** **p-values** later
- Probability describes outcomes **before** we see data

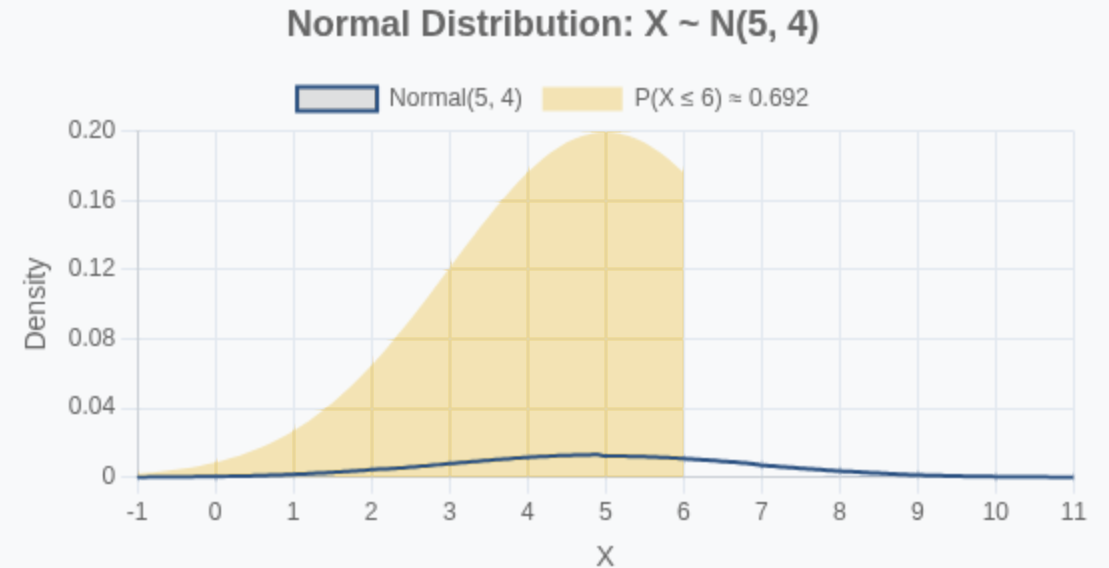
**Notation:**  $f(x)$  represents PDF (density function),  $F(x)$  represents CDF (cumulative distribution)

## Practice: Appendix B.2 (Normal probabilities)

If  $X \sim \text{Normal}(5, 4)$ , compute:

- (i)  $P(X \leq 6)$
- (ii)  $P(X > 4)$
- (iii)  $P(|X - 5| > 1)$

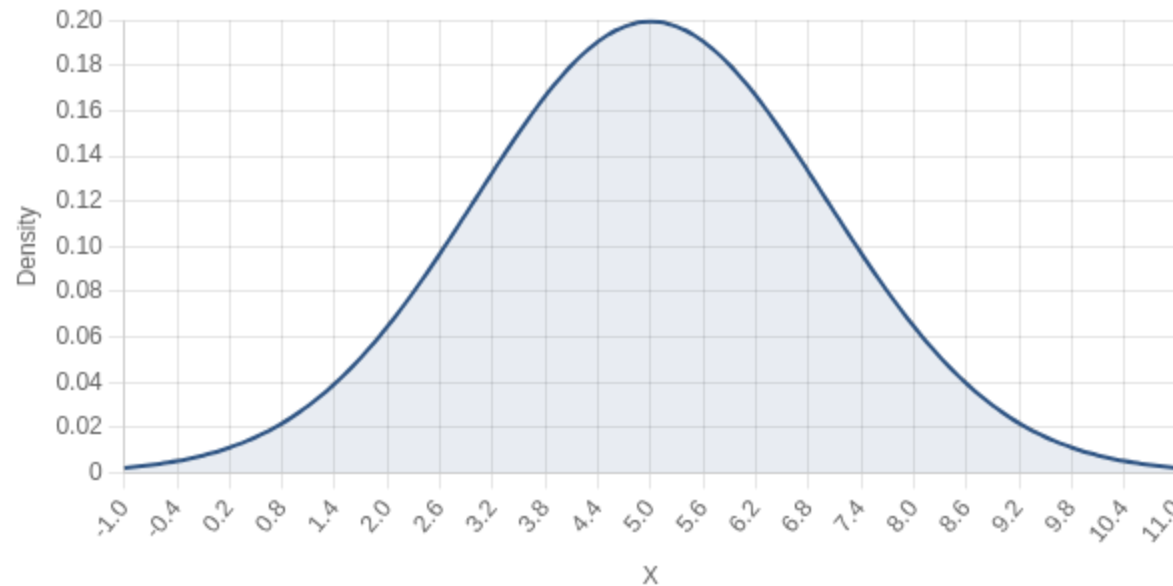
*Solution will follow on a later slide...*



# Normal Probability Calculations

## Appendix B.2 (Normal probabilities)

If  $X \sim \text{Normal}(5, 4)$ , compute: (i)  $P(X \leq 6)$ , (ii)  $P(X > 4)$ , (iii)  $P(|X - 5| > 1)$ .



- **Step 1: Standardize** with  $Z = (X - 5)/2$  where  $\mu = 5$ ,  $\sigma = 2$ .
- **Step 2: For  $P(X \leq 6)$**   $Z = (6 - 5)/2 = 0.5$   $P(X \leq 6) = P(Z \leq 0.5) \approx 0.692$
- **Step 3: For  $P(X > 4)$**   $Z = (4 - 5)/2 = -0.5$   $P(X > 4) = P(Z > -0.5) = 1 - P(Z \leq -0.5) = 1 - 0.308 \approx 0.692$
- **Step 4: For  $P(|X-5| > 1)$**   $P(|X-5| > 1) = P(X < 4) + P(X > 6) = P(Z < -0.5) + P(Z > 0.5) = 0.308 + 0.308 = 0.616$

### Final Answers:

(i)  $P(X \leq 6) = 0.692$

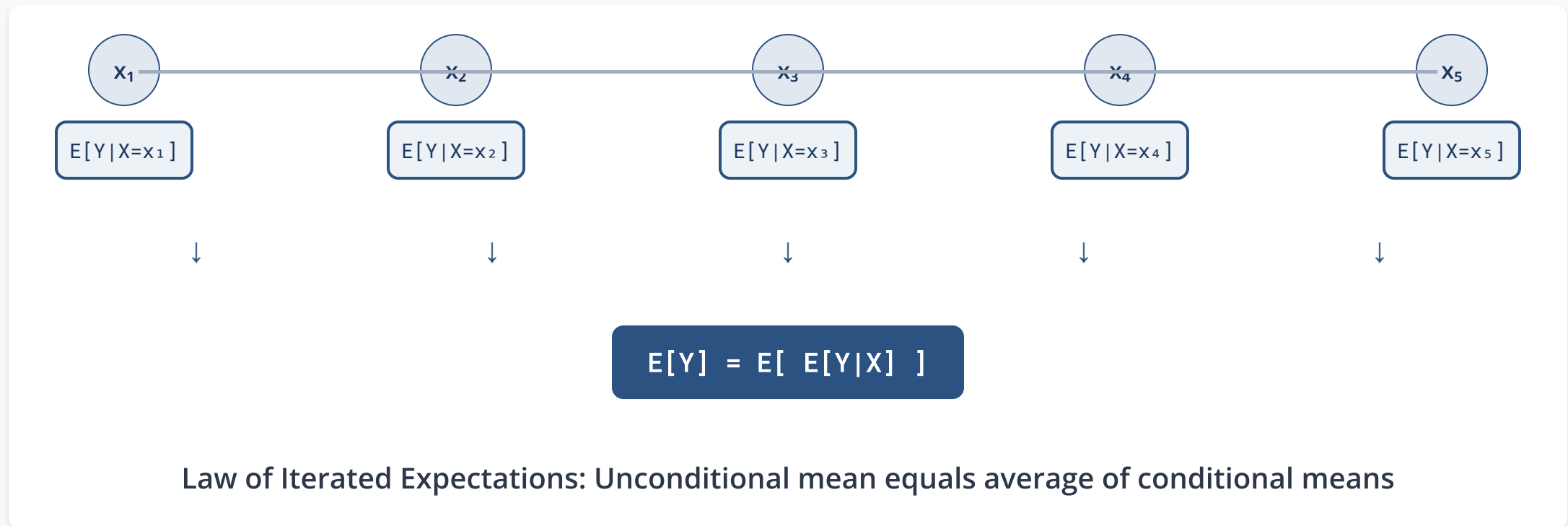
(ii)  $P(X > 4) = 0.692$

(iii)  $P(|X - 5| > 1) = 0.616$



# Conditional expectation & LIE (bridge to regression)

- $E[Y|X]$  as best predictor of  $Y$  given  $X$  — **population regression function**
- Law of Iterated Expectations links conditional and unconditional means
- $E[Y|X]$  varies with  $X$ ,  $E[Y]$  is the average of these conditional means



## Key LIE Properties:

$E[Y] = E[ E[Y|X] ]$  — "Averaging over averages"

$E[g(X)Y|X] = g(X)E[Y|X]$  — "Pull out known functions"

$E[Y|X] = E[Y]$  when  $X$  and  $Y$  are independent

# Conditional Expectation: Practice — Appendix B.10

## Practice Problem

Given  $E(\text{GPA} \mid \text{SAT}) = 0.70 + 0.002 \cdot \text{SAT}$  and  $E(\text{SAT}) = 1100$ :

- (i) Find  $E(\text{GPA} \mid \text{SAT}=800)$  and  $E(\text{GPA} \mid \text{SAT}=1400)$
- (ii) Find  $E(\text{GPA})$  using Law of Iterated Expectations

### Law of Iterated Expectations (LIE)

$$E(\text{GPA} \mid \text{SAT}) = 0.70 + 0.002 \cdot \text{SAT} \text{ Conditional Expectation}$$



$$E[\text{GPA}] = E[E(\text{GPA} \mid \text{SAT})] \text{ Law of Iterated Expectations}$$



$$E[\text{GPA}] = E[0.70 + 0.002 \cdot \text{SAT}] \text{ Substitute the conditional expectation}$$

## Solution Steps

- 1 For  $E(\text{GPA} \mid \text{SAT}=800)$ : Substitute 800 into conditional mean formula  
 $E(\text{GPA} \mid \text{SAT}=800) = 0.70 + 0.002(800) = 0.70 + 1.60 = 2.30$
- 2 For  $E(\text{GPA} \mid \text{SAT}=1400)$ : Substitute 1400 into conditional mean formula  
 $E(\text{GPA} \mid \text{SAT}=1400) = 0.70 + 0.002(1400) = 0.70 + 2.80 = 3.50$
- 3 For  $E(\text{GPA})$ : Apply LIE and linearity of expectation  
 $E[\text{GPA}] = E[0.70 + 0.002 \cdot \text{SAT}] = 0.70 + 0.002 \cdot E[\text{SAT}] = 0.70 + 0.002(1100) = 2.90$

## Answer

- (i)  $E(\text{GPA} \mid \text{SAT}=800) = 2.30$ ,  $E(\text{GPA} \mid \text{SAT}=1400) = 3.50$
- (ii)  $E[\text{GPA}] = 2.90$

# From Probability to Sampling

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Now we transition from foundational probability concepts to sampling distributions and estimation — the bridge between theory and statistical inference



Sampling  
Distributions

Sample Mean  
Properties

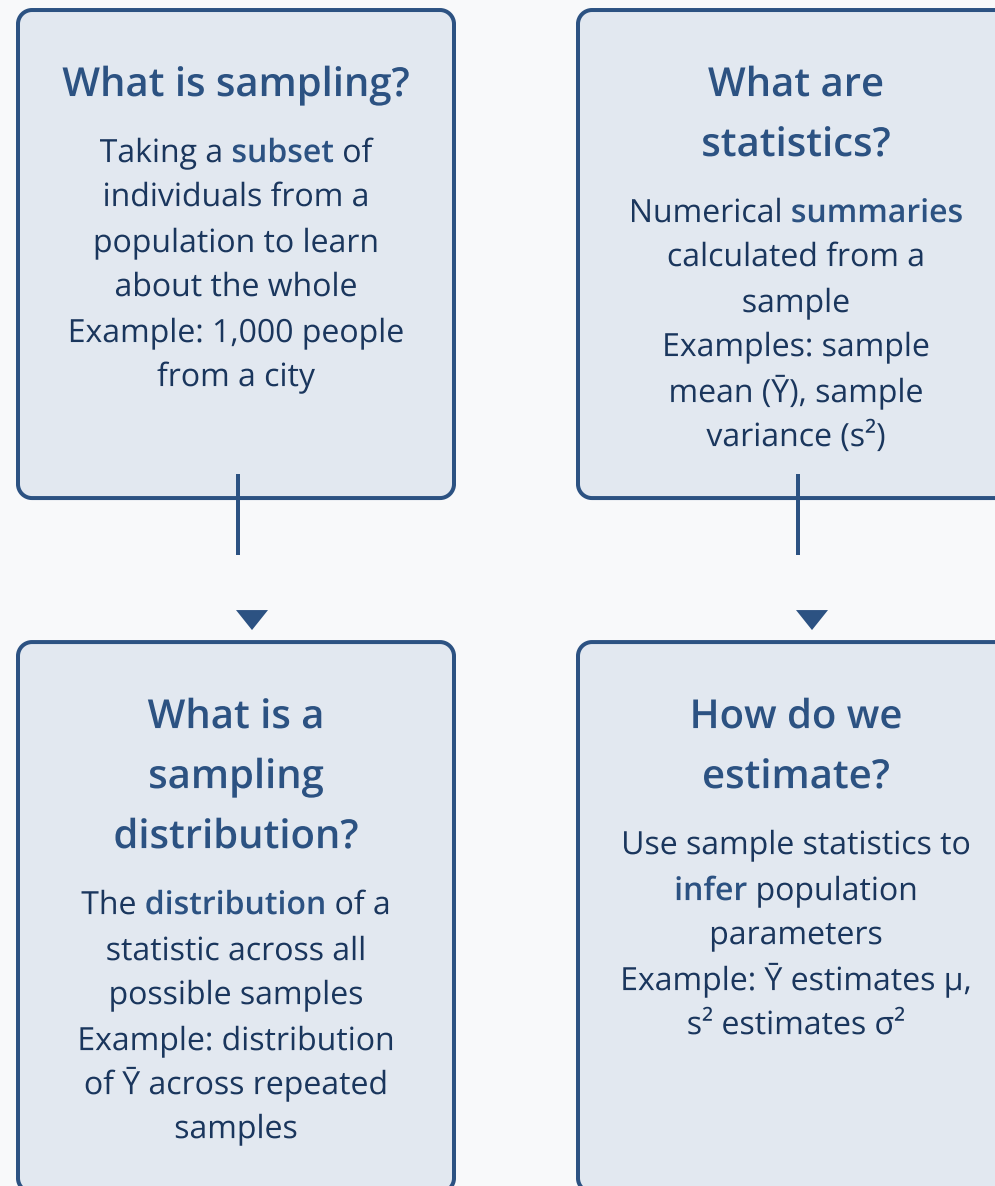
Variance  
Reduction

Statistical  
Inference

# Key Definitions: From Sampling to Estimation

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Understanding the progression from sampling to statistical inference



**Key insight: The sampling distribution connects individual samples to population parameters**

This is why we need the Central Limit Theorem and Law of Large Numbers

# Sampling vs. Sampling Distribution

- **Sampling:** Taking individual observations from a population
- **Sampling distribution:** How statistics like  $\bar{Y}$  vary if we repeated sampling
- Standard errors quantify this hypothetical variation

## Sampling

Taking a specific set of  $n$  observations from the population

## Sampling Distribution

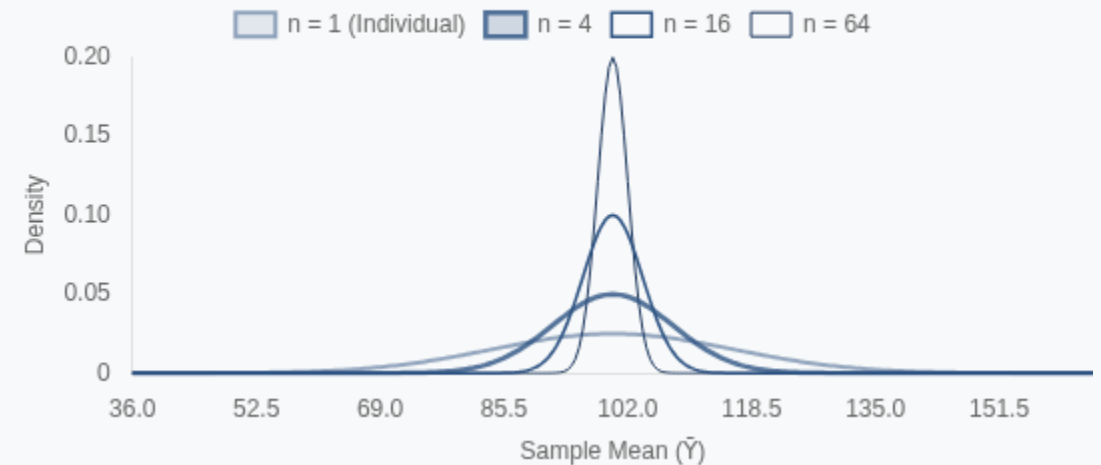
The distribution of sample statistics across all possible samples

## Key Insight

The standard error of the mean ( $SE = \sigma/\sqrt{n}$ ) measures the standard deviation of the sampling distribution—not the sample itself!

This is why we can make inferences about population parameters using just one sample.

Sampling Distribution Narrows as  $n$  Increases



# Sampling Mean Properties — Appendix C.1

## Practice Problem

For  $n = 4$  i.i.d. draws with mean  $\mu$  and variance  $\sigma^2$ , show:

- (i)  $E(\bar{Y}) = \mu$
- (ii)  $\text{Var}(\bar{Y}) = \sigma^2/4$

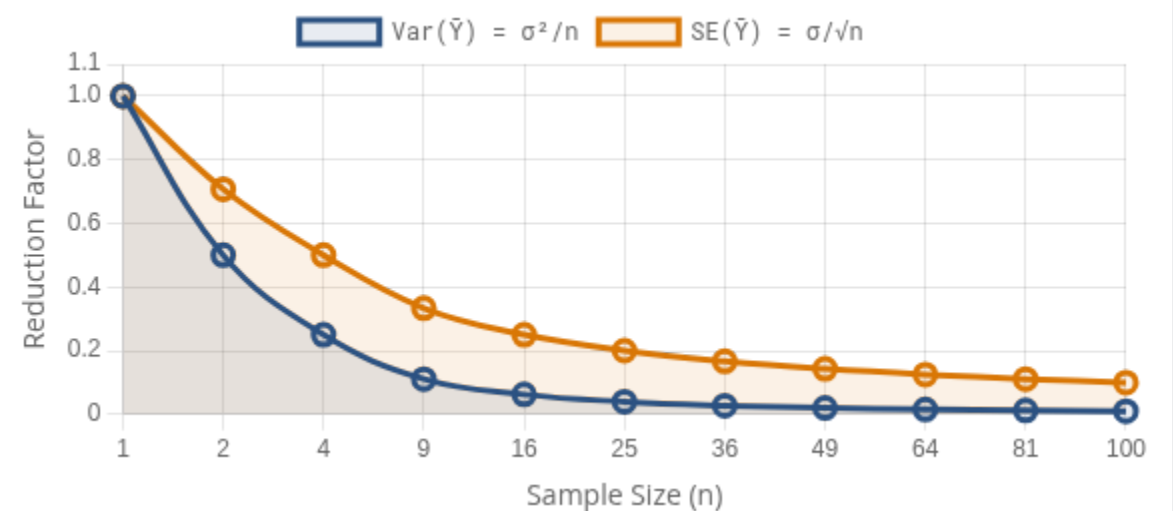
## Solution Steps

- 1 Express the sample mean:  $\bar{Y} = (X_1 + X_2 + X_3 + X_4)/4$
- 2 Use **linearity of expectation**:  
 $E(\bar{Y}) = E((X_1 + X_2 + X_3 + X_4)/4) = (E(X_1) + E(X_2) + E(X_3) + E(X_4))/4$
- 3 Since all draws are identical:  $E(X_1) = E(X_2) = E(X_3) = E(X_4) = \mu$
- 4 Therefore:  $E(\bar{Y}) = (\mu + \mu + \mu + \mu)/4 = \mu$  (**unbiased**)
- 5 For variance, use **independence**:  
 $\text{Var}(\bar{Y}) = \text{Var}((X_1 + X_2 + X_3 + X_4)/4) = (1/16)(\text{Var}(X_1 + X_2 + X_3 + X_4))$
- 6 By independence:  $\text{Var}(X_1 + X_2 + X_3 + X_4) = \text{Var}(X_1) + \text{Var}(X_2) + \text{Var}(X_3) + \text{Var}(X_4)$
- 7 Since all draws have the same variance:  $\text{Var}(X_1) = \text{Var}(X_2) = \text{Var}(X_3) = \text{Var}(X_4) = \sigma^2$
- 8 Therefore:  $\text{Var}(\bar{Y}) = (1/16)(4\sigma^2) = \sigma^2/4 = \sigma^2/n$

## Answer

- (i)  $E(\bar{Y}) = \mu$
- (ii)  $\text{Var}(\bar{Y}) = \sigma^2/4 = \sigma^2/n$

## Variance Reduction Effect of Averaging



## Why This Matters

- **Unbiasedness:** Sample mean targets the true mean
- **Variance reduction:** More data = less uncertainty
- **Standard error:**  $\text{SE}(\bar{Y}) = \sigma/\sqrt{n}$  (decreases with sample size)

## Connection to CLT

This variance formula powers the Central Limit Theorem:

$\bar{Y} \sim \text{Normal}(\mu, \sigma^2/n)$  for large  $n$

This enables all of our t-tests and confidence intervals!

# LLN: why averages stabilize

- $\bar{Y} \rightarrow \mu$  in probability (**consistency**)

As  $n$  grows, sample mean approaches population mean

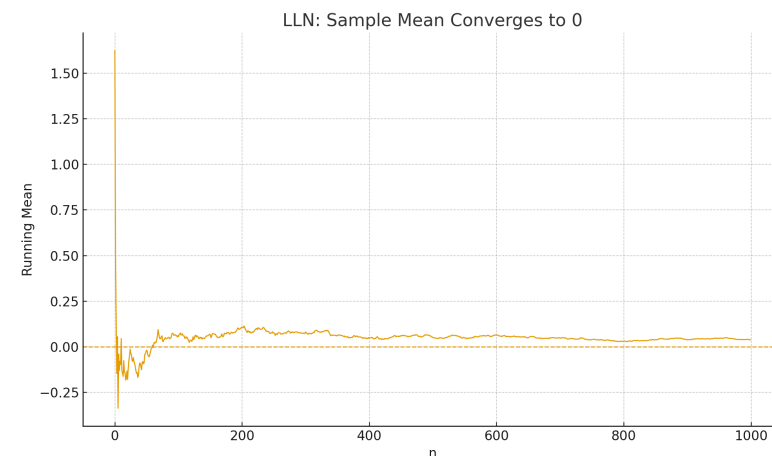
- **Variance reduction** drives convergence

$\text{Var}(\bar{Y}) = \sigma^2/n \rightarrow 0$  as  $n$  increases

- Connects to **sampling distribution**

Remember: Sampling variance shrinks with sample size

## LLN: Sample Mean Converges to $\mu$



## From LLN to Econometrics

### Sampling $\rightarrow$ LLN

The sample mean's variance shrinks as  $n$  increases, making estimates more precise

### LLN $\rightarrow$ Consistency

OLS estimators are consistent because they converge to true population parameters as  $n \rightarrow \infty$

### Why It Matters

With large enough samples, we can be confident our estimates are close to the truth

# LLN Practice — Appendix C.1 with Chebyshev

## Practice Problem

Let  $Y_1, \dots, Y_n$  be i.i.d. with  $E(Y_i) = \mu$  and  $\text{Var}(Y_i) = \sigma^2 < \infty$ . Using C.1 and Chebyshev, show that for any  $\varepsilon > 0$ :

$$P(|\bar{Y} - \mu| \geq \varepsilon) \leq \sigma^2 / (n\varepsilon^2) \rightarrow 0 \text{ as } n \rightarrow \infty \text{ (i.e., an LLN).}$$

## Solution Steps

- 1 From Appendix C.1, recall that for the sample mean  $\bar{Y}$ :  
 $E(\bar{Y}) = \mu$  and  $\text{Var}(\bar{Y}) = \sigma^2/n$
- 2 Apply Chebyshev's inequality, which states:  
 $P(|X - E(X)| \geq \varepsilon) \leq \text{Var}(X)/\varepsilon^2$
- 3 Substitute  $\bar{Y}$  for  $X$ ,  $\mu$  for  $E(X)$ , and  $\sigma^2/n$  for  $\text{Var}(X)$ :  
 $P(|\bar{Y} - \mu| \geq \varepsilon) \leq \text{Var}(\bar{Y})/\varepsilon^2 = (\sigma^2/n)/\varepsilon^2 = \sigma^2/(n\varepsilon^2)$
- 4 As  $n \rightarrow \infty$ ,  $\sigma^2/(n\varepsilon^2) \rightarrow 0$  for any fixed  $\varepsilon > 0$  and  $\sigma^2 < \infty$ .  
Therefore,  $P(|\bar{Y} - \mu| \geq \varepsilon) \rightarrow 0$ , which means  $\bar{Y} \rightarrow^p \mu$

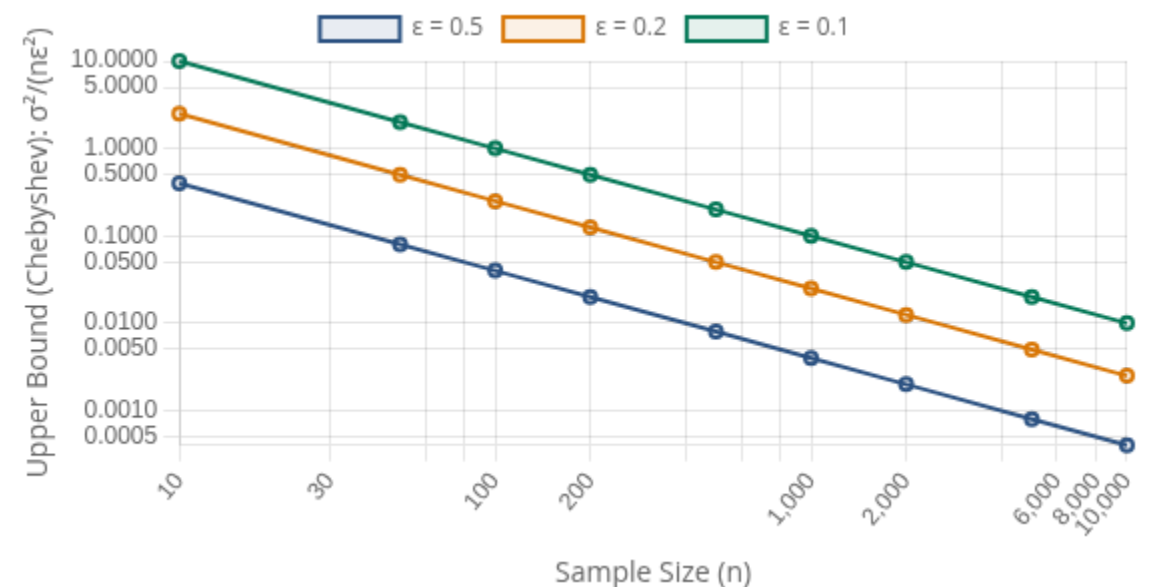
## Answer

$P(|\bar{Y} - \mu| \geq \varepsilon) \leq \sigma^2 / (n\varepsilon^2)$  (by Chebyshev)  
Since  $\sigma^2 / (n\varepsilon^2) \rightarrow 0$  as  $n \rightarrow \infty$ , we have  $\bar{Y} \rightarrow^p \mu$  (Weak LLN)

## Key Insight

This is one of the most elegant proofs of the Weak Law of Large Numbers, requiring only Chebyshev's inequality and the basic properties of sample means. It shows why we need only the first two moments (mean and variance) to establish consistency.

## Visualizing Convergence in Probability



## Chebyshev's Bound for Different n Values

For  $n = 10$ :  $P(|\bar{Y} - \mu| \geq 0.5) \leq \sigma^2 / (10 \cdot 0.5^2) = \sigma^2 / 2.5$   
For  $n = 100$ :  $P(|\bar{Y} - \mu| \geq 0.5) \leq \sigma^2 / (100 \cdot 0.5^2) = \sigma^2 / 25$   
For  $n = 1000$ :  $P(|\bar{Y} - \mu| \geq 0.5) \leq \sigma^2 / (1000 \cdot 0.5^2) = \sigma^2 / 250$



# Hypothesis Testing & Inference

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Moving from estimation to interpretation — how do we determine if patterns are real or just statistical noise?



Null Hypothesis

P-values

Significance  
Levels

Confidence  
Intervals

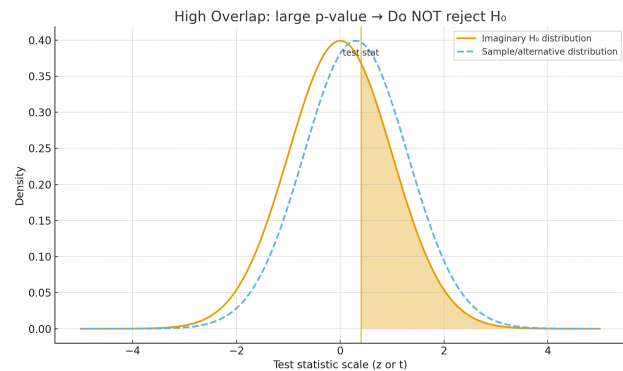
# Understanding Hypothesis Testing Distributions

## What distribution are we actually looking at?

In hypothesis testing, we're examining where our **test statistic** lands within an **imaginary distribution** that would exist if the null hypothesis ( $H_0$ ) were true.

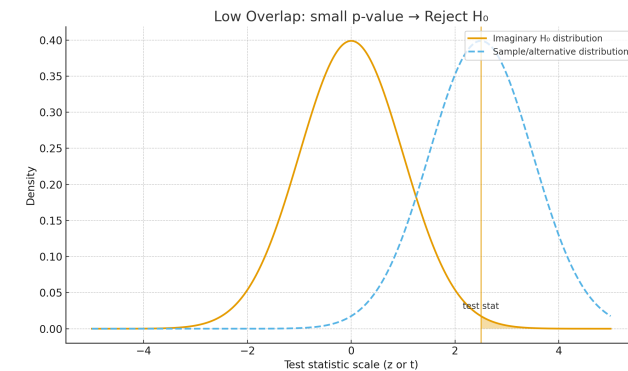
- 1 We have an **imaginary true distribution** based on  $H_0$  (what the world would look like if null is true)
- 2 We examine where our **test statistic** (what we actually observed) lands in this imaginary distribution
- 3 If our test statistic is too **extreme** (unlikely under  $H_0$ ), we reject  $H_0$  in favor of our alternative

### High Overlap Scenario



Large p-value → Don't Reject  $H_0$   
(Sample result could easily happen under  $H_0$ )

### Low Overlap Scenario



Small p-value → Reject  $H_0$   
(Sample result would be rare under  $H_0$ )

# Hypothesis Testing: The Big Idea

*"If there were truly no effect (the null hypothesis,  $H_0$ ), how weird is our data?"*

## Test statistic:

A single number showing 'how many standard errors away' from what  $H_0$  predicts. Larger magnitude = more unusual.

## p-value:

Probability, assuming  $H_0$  is true, of getting a result at least as extreme as observed. Small  $p$  = rare under  $H_0$ .

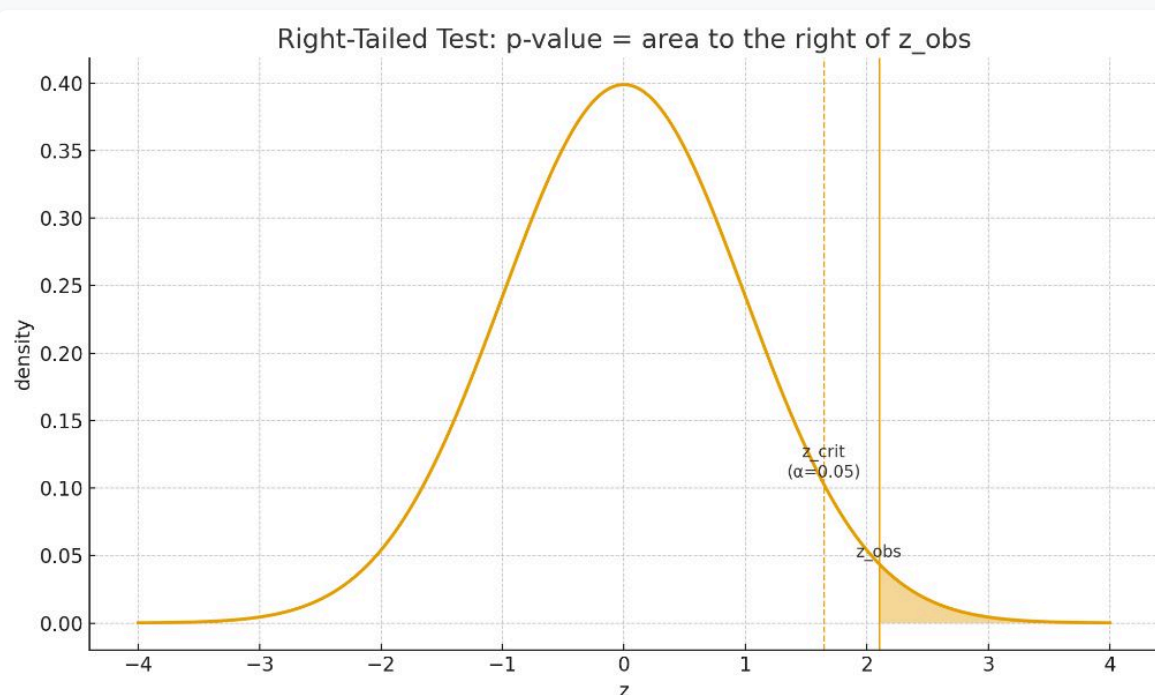
## Critical value / $\alpha$ (alpha):

A pre-set risk tolerance (e.g.,  $\alpha=0.05$ ) marking a cutoff on the curve. Reject  $H_0$  if test statistic passes this line.

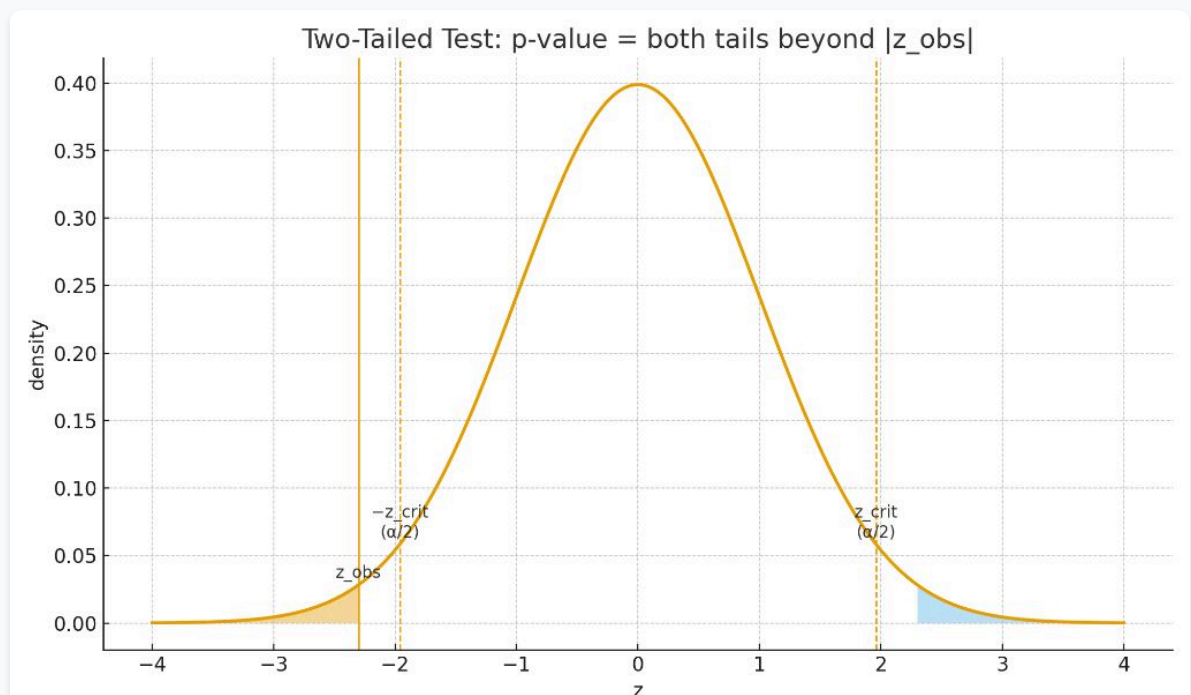
## Rejection meaning:

"This sample would be unlikely if  $H_0$  were true." Not that  $H_0$  is impossible, but inconsistent with data at the chosen error rate.

### Right-Tailed Test: p-value = area to the right of $z_{\text{obs}}$



### Two-Tailed Test: p-value = both tails beyond $|z_{\text{obs}}|$



## Two equivalent decision rules:

### p-value rule:

Reject  $H_0$  if  $p \leq \alpha$ .

### Critical-value rule:

Reject  $H_0$  if  $|\text{test stat}| \geq \text{critical value}$ .

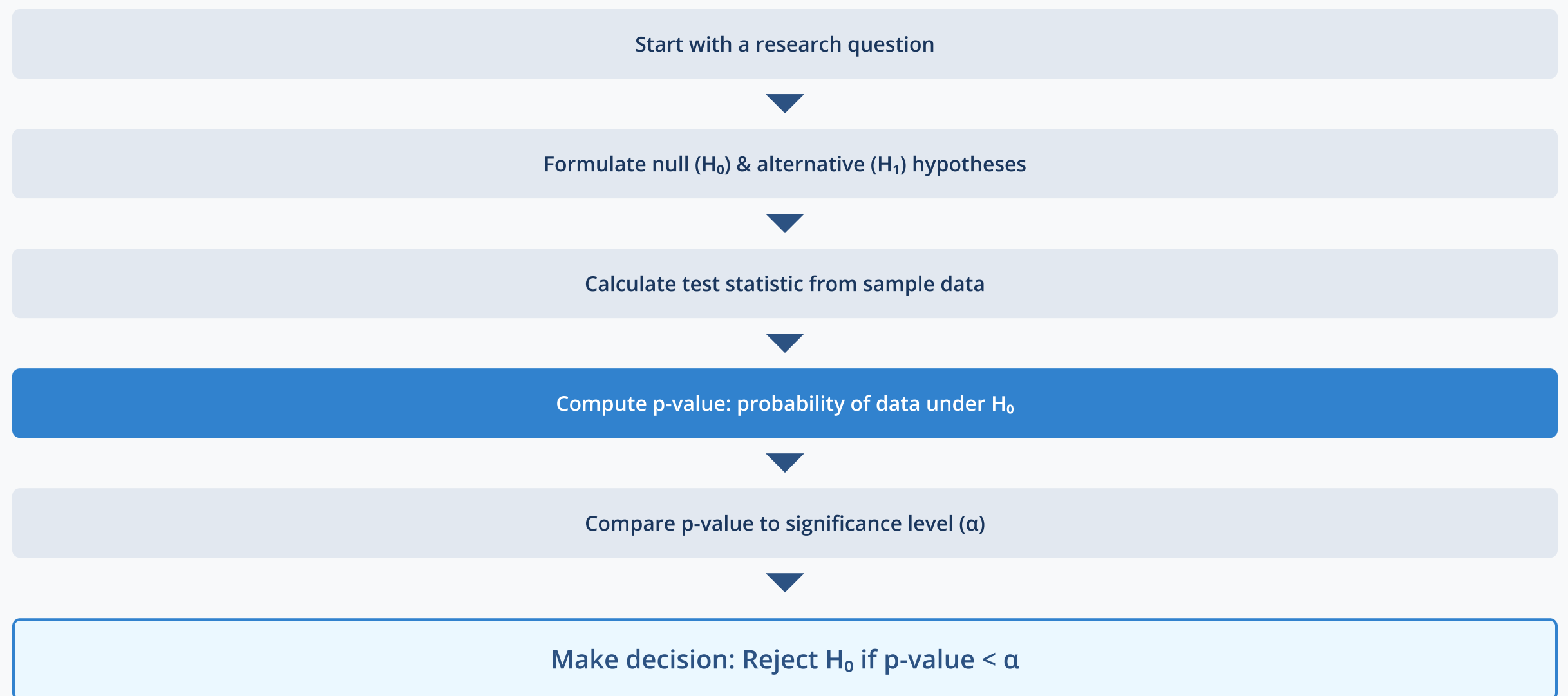
### Type I Error:

False alarm - rejecting a true  $H_0$ . Occurs at rate  $\alpha$ .

### Type II Error:

Missed effect - failing to reject false  $H_0$ . Controlled by power and sample size.

# Key Definitions: Hypothesis Testing Concepts



## Hypothesis

A statement about a population parameter that can be tested with sample data

## Null Hypothesis (H<sub>0</sub>)

The "status quo" claim we're testing against; typically states "no effect" or "no difference" (e.g.,  $\beta = 0$ )

## Alternative Hypothesis (H<sub>1</sub>)

The claim we suspect is true instead of H<sub>0</sub>; what we're looking for evidence to support

## p-value

Probability of observing data at least as extreme as our sample, assuming H<sub>0</sub> is true; smaller p-values indicate stronger evidence against H<sub>0</sub>

## Significance Level (α)

Pre-determined threshold for rejection (typically 0.01, 0.05, or 0.10); represents acceptable Type I error risk

## Type I vs Type II Errors

Type I: Rejecting a true H<sub>0</sub> (false positive)  
Type II: Failing to reject a false H<sub>0</sub> (false negative)

# Hypothesis Testing: Practice — Appendix C.6

## Practice Problem

Given  $\bar{y} = -32.8$ ,  $s = 466.4$ ,  $n = 30$

Test  $H_0: \mu=0$  against  $H_1: \mu<0$  (one-sided)

Compute  $t$  and the  $p$ -value

## Solution Steps

- 1 Calculate standard error:  $se(\bar{y}) = s/\sqrt{n} = 466.4/\sqrt{30}$

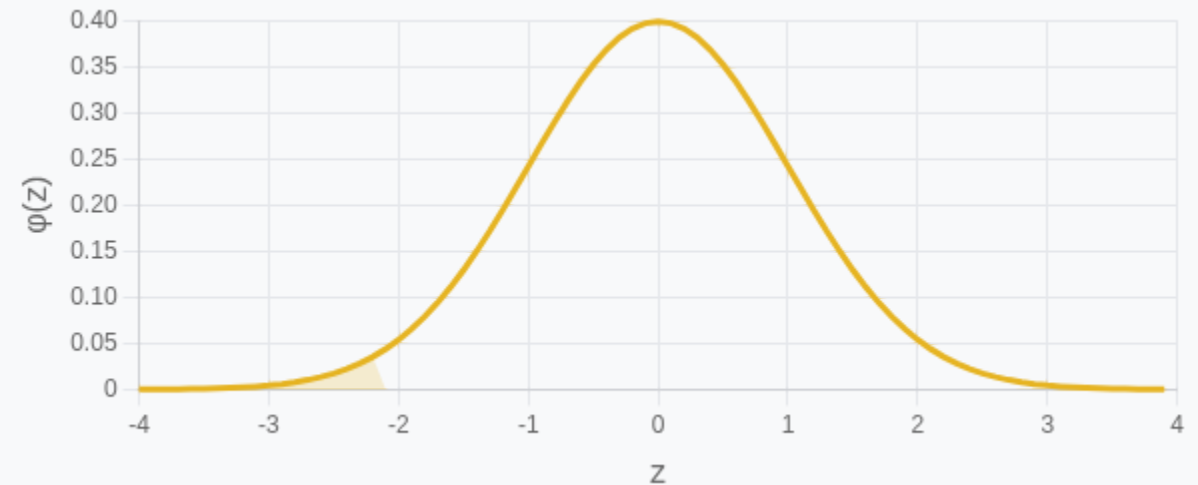
$$se(\bar{y}) = 466.4/\sqrt{30} = 466.4/5.477 \approx 15.55$$

- 2 Calculate  $t$ -statistic:  $t = (\bar{y} - \mu_0) / se(\bar{y})$

$$t = (-32.8 - 0)/15.55 \approx -2.11$$

- 3 Find one-sided  $p$ -value from Normal(0,1) table

$$p = P(Z \leq -2.11) \approx 0.017$$



One-sided  $p$ -value ( $z \leq -2.11$ ): area  $\approx 0.017$

## Complete Answer

$$se(\bar{y}) = 466.4/\sqrt{30} \approx 15.55$$

$$t = -32.8/15.55 \approx -2.11$$

$$p \approx 0.017$$

→ reject  $H_0$  at 5% significance level

→ do not reject at 1% level

## Decision Rule & Interpretation

Since  $p \approx 0.017 < 0.05$ , we reject  $H_0$  at the 5% significance level.

However, since  $p > 0.01$ , we would not reject at the 1% level.

We conclude there is statistically significant evidence that  $\mu < 0$ .

# Matrix algebra I: transpose & symmetry

- Matrix notation compresses regression:  $y = X\beta + u$   
and  $\hat{\beta} = (X'X)^{-1}X'y$
- A basic but key fact:  $X'X$  is symmetric
- Symmetry underlies OLS properties and numerical stability

$$\boxed{X'X} \rightarrow \boxed{(X'X)'} \rightarrow \boxed{X'X}$$

$k \times k$  symmetric

## Appendix D.3 (transpose identity)

Show that  $(X'X)' = X'X$ .

1. Use  $(AB)' = B'A'$
2. Apply to  $A=X'$  and  $B=X$
3. Use  $(A')' = A$

Direct application yields  $X'X$ .

# Matrix Algebra II & III: Inverses & Positive Definiteness

"When inverting a product, the order reverses—used constantly in derivations. Positive definite (PD) means  $x'Ax > 0$  for any nonzero  $x$ . If  $X$  has full column rank,  $X'X$  is PD—hence invertible and OLS is unique."

## D.5: Inverse of a Product

Verify  $B^{-1}A^{-1}$  is the inverse of  $AB$ ; conclude  $(ABC)^{-1} = C^{-1}B^{-1}A^{-1}$ .

Check left inverse:  $C(AB) = B^{-1}A^{-1}AB = B^{-1}IB = B^{-1}B = I$

Check right inverse:  $(AB)C = ABB^{-1}A^{-1} = AIA^{-1} = AA^{-1} = I$

Since both  $C(AB) = I$  and  $(AB)C = I$ ,  $C$  is indeed the inverse of  $AB$ .

**Conclude:  $(ABC)^{-1} = C^{-1}B^{-1}A^{-1}$  — order always reverses!**

## D.6: PD Matrix Diagonals

Show diagonal elements of a positive definite matrix are strictly positive.

Let  $A$  be PD, so  $x'Ax > 0$  for all  $x \neq 0$

Choose  $x = e_j$  (unit vector with 1 in position  $j$ , 0 elsewhere)

Then  $e_j'Ae_j = a_{jj} > 0$

The  $j$ th diagonal entry equals the quadratic form when using the  $j$ th unit vector, which must be positive by the PD property.

## D.7: Congruence Preserves PD

If  $A$  is PD and  $P$  is invertible, show  $P'AP$  is positive definite.

For any  $x \neq 0$ :

Let  $z = Px$  (note  $z \neq 0$  since  $P$  is invertible)

Then  $x'(P'AP)x = (Px)'A(Px) = z'Az > 0$

Since  $P$  is invertible,  $z = Px$  is nonzero whenever  $x$  is nonzero, and  $z'Az > 0$  by the PD property of  $A$ . Hence  $P'AP$  satisfies the definition of positive definiteness.

## Why This Matters for OLS

If  $X$  has full column rank, why is OLS solution unique?

Full column rank  $\rightarrow X'X$  is PD  $\rightarrow X'X$  is invertible

$\rightarrow \hat{\beta} = (X'X)^{-1}X'y$  has unique solution

**These properties ensure  $\hat{\beta}$  exists and is unique—essential for regression inference!**

# Putting it together: from stats to econometrics

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Probability  
→ targets

Sampling & CLT  
→ behavior

Testing & Matrices  
→ decisions & computation

- Probability **defines** the population targets we aim to estimate
- Sampling and CLT **explain** why our estimates are well-behaved
- Tests and matrix tools **enable** practical decisions and efficient computation

*"This statistical spine supports every regression table you'll encounter. Keep this framework in mind as we move to OLS and beyond."*



# What's next

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- Simple regression: **model, assumptions, OLS, interpretation**
- Uncertainty first; **causality with identification** later
- Keep today's **statistical spine** in mind when we move to OLS